

Image Enhancement Report Analysis

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PCD Assignment 02

Objective and Experimental Design

This experiment evaluates four enhancement techniques for dark, overly bright, low-contrast, and blurred images. The uploaded file pcd02.jfif provides a single RGB reference at 960×1280 pixels. The notebook saves its processed working copy as images/reference.png. Each degraded input is generated independently from this reference, so the same image can be used for a controlled before-and-after comparison.

The photograph shows an outdoor school setting with buildings, palm trees, and a group of people. It contains both bright sky and shaded architectural areas, along with visible edges and textures. These features make changes in tone and sharpness apparent during visual inspection. The report presents measurements and descriptions without reproducing the photograph.

Image Degradation and Enhancement

Condition	Generated input	Enhancement
Dark	Gamma 1.55	Gamma 0.65
Bright	Gamma 0.69	Gamma 1.45
Low contrast	Pixel values $\times 0.52 + 61$	Auto-contrast, cut off 1%
Blurred	Gaussian blur, radius 1.5	Unsharp mask, radius 1.5, strength 200%, threshold 0

Gamma correction changes midtone brightness. A gamma above 1 darkens the generated input, a gamma below 1 brightens it. For low contrast, the notebook compresses the RGB range and then expands it using automatic contrast stretching. Gaussian blur reduces local edge detail, unsharp masking emphasizes the edges remaining in the blurred input.

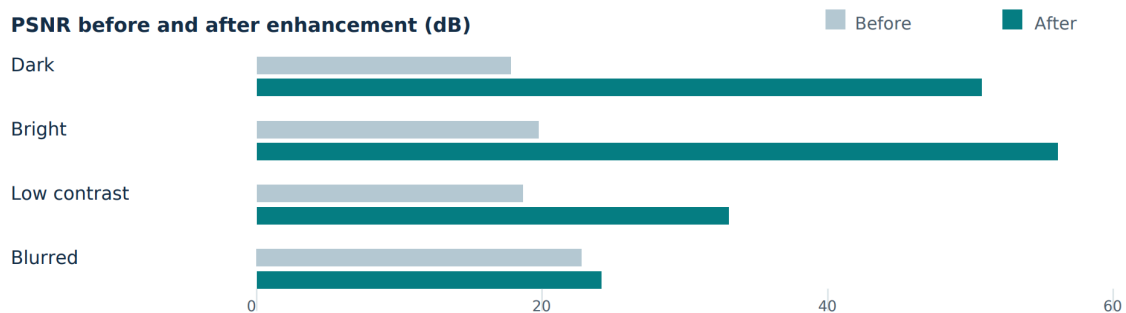
Evaluation Procedure

Three images within a case have the same dimensions and three color channels. For each degraded input and enhanced output, the notebook calculates the mean squared error (MSE) against the working reference over all RGB values, then computes $PSNR = 10 \log_{10}(255^2 / MSE)$ in decibels. A higher PSNR indicates lower numerical error relative to that reference. The comparison figure is inspected separately because PSNR alone does not fully capture perceived visual quality.

Quantitative Results

Condition	Before (dB)	After (dB)	Change (dB)
Dark	17.80	50.76	+32.95
Bright	19.74	56.09	+36.34
Low contrast	18.65	33.04	+14.39
Blurred	22.78	24.10	+1.32

PSNR before and after enhancement (dB)



Visual interpretation

- Dark and bright: The corrected outputs look close to the reference in the comparison figure. Their large PSNR gains are expected because the selected correction gammas are close to the mathematical inverses of the simulated degradation gammas. The figures demonstrate this chosen simulation rather than proving comparable restoration for an unrelated underexposed or overexposed camera image.
- Low contrast: The compressed input appears flatter. Automatic contrast stretching makes tonal separation more apparent and increases PSNR from 18.65 to 33.04 dB. Some difference from the reference remains because compression and integer rounding discard distinctions between pixel values.
- Blurred: Unsharp masking restores some edge definition, but the output remains softer than the reference. Its improvement of 1.32 dB is much smaller than for the tonal conditions. Blurring removes fine spatial detail, emphasizing surviving edges cannot recreate all original textures.

Limitations and conclusion

PSNR measures pixel-wise similarity and is only meaningful here because the undamaged reference is available. It does not by itself measure whether an output looks natural, and it should not be used as a ground-truth score for a separate real-world photo with no matching reference. The four degradations are deliberately controlled and parameter-specific. All four methods raised PSNR in this run, with the largest measured gain for brightness correction and the smallest for blur. A future extension could evaluate additional photographs, stronger degradations, and perceptual metrics alongside visual inspection.